

Smart Home Energy Forecasting: Visual Analysis Report

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1. Executive Summary & Overview

This report provides a comprehensive qualitative and quantitative explanation of the visual analytical plots generated during the Exploratory Data Analysis (EDA), dimensionality reduction (PCA), unsupervised anomaly clustering (KMeans), supervised model benchmarking (Regression & Classification), and time series forecasting (ARIMA) phases of the Smart Home Energy Analytics system.

Plot 1: Distribution of Total Energy Consumption (Total_Energy_kWh)

Analyzes the right-skewed energy consumption profile across households. Indicates baseline standby energy draw around 1-3 kWh with peak usage tailing up to 15+ kWh during heavy HVAC or concurrent appliance usage periods.

Plot 2: Energy Consumption Variation Across Home IDs

Displays boxplots comparing energy metrics across individual smart homes. Highlights baseline house-level variance influenced by home footprint size, occupants count, and thermal efficiency ratings.

Plot 3: Energy Consumption Distribution by Hour of Day

Reveals strong daily diurnal patterns. Night hours (00:00 - 05:00) demonstrate minimal baseline consumption, while morning (07:00 - 09:00) and evening peak hours (18:00 - 21:00) exhibit maximum variance and peak load spikes.

Plot 4: Energy Consumption Comparison by Day of the Week

Compares weekday versus weekend consumption profiles. Weekends show prolonged mid-day energy draw due to higher occupant presence and daytime appliance usage.

Plot 5: Monthly Average Energy Consumption Trends

Demonstrates long-term seasonal trends, peaking during summer/winter extremes due to elevated heating and air-conditioning (HVAC) demands.

Plot 6: Hourly Average Load Profile

Provides a continuous line plot of average load progression across 24-hour cycles, serving as a key benchmark input for predictive load-shaping models.

Plot 7: Ambient Temperature vs. Total Energy Consumption Scatter

Exhibits a characteristic U-shaped non-linear relationship where extreme low (heating demand) and extreme high (cooling demand) temperatures trigger significant increases in Total Energy consumption.

Plot 8: Feature Correlation Heatmap

Measures Pearson correlation coefficients between environmental (temperature, humidity), temporal (hour, day, holiday), and appliance-level loads (HVAC, Fridge, Washing Machine, Lighting). Identifies HVAC as the dominant energy driver.

Plot 9: Appliance-Level Energy Contribution Breakdown

Categorizes total energy distribution by individual appliances, establishing that HVAC systems account for ~45% of total power draw, followed by refrigeration, washing machines, and lighting.

Plot 10: Multi-Feature Pairplot Matrix

Illustrates pairwise joint distributions and feature interactions across primary continuous predictors, confirming linear and polynomial interaction terms for model feature engineering.

Plot 11: PCA Scree Plot (Explained Variance)

Plots cumulative explained variance ratios across principal components. Shows that the first 2 Principal Components capture >78% of the total dataset variance.

Plot 12: 2D Principal Component Projection (PC1 vs PC2)

Projects 100,000 high-dimensional feature points onto a 2D subspace, displaying clear spatial separation corresponding to low, moderate, and high load operational regimes.

Plot 13: KMeans Clustering Elbow Curve

Evaluates Within-Cluster Sum of Squares (WCSS) across cluster values $K=1..10$. Demonstrates an optimal 'elbow' at $K=3$, establishing distinct operational clusters: Normal, High Load, and Energy Waste.

Plot 14: KMeans Anomaly & Waste Cluster PCA Projection

Visualizes the 3 discovered clusters in PCA space. Highlights Cluster 2 as anomalous 'Energy Waste' characterized by high energy draw during low occupant activity or mild ambient temperatures.

Plot 15: Waste Profile Boxplot Distribution

Compares feature value distributions across normal versus wasteful clusters, proving that waste incidents are predominantly caused by unoptimized HVAC schedules and prolonged idle appliance activation.

Plot 16: Supervised Regression Model Comparison (Linear, RF, XGBoost)

Compares model metrics (RMSE, MAE, R^2 score). Shows XGBoost Regressor achieving superior predictive performance ($R^2 > 0.94$) for continuous energy forecasting.

Plot 17: Supervised Waste Classifier Benchmarking (Precision, Recall, F1-Score)

Evaluates Random Forest and XGBoost classifiers on identifying energy waste instances. XGBoost achieves >95% F1-score and high ROC-AUC.

Plot 18: ARIMA Time Series Model Fit & Residual Analysis

Displays historical actuals against ARIMA fitted values, validating stationarity (ADF test $p < 0.05$) and residual white noise properties for temporal forecasting.

Plot 19: 72-Hour Ahead Energy Demand Forecast with Confidence Intervals

Presents multi-step out-of-sample time series forecasts with 95% prediction intervals, demonstrating effective peak demand projection for smart grid load balancing.

Cleanup Note: Per maintainer instructions, all intermediate PNG image files have been compiled into this comprehensive document and removed from the codebase repository.